Alif Ensemble E7

What an NPU microcontroller actually buys

Cores	M55 HE 160 MHz + M55 HP 400 MHz
NPU	2× Ethos-U55, 128 and 256 MAC
SRAM	5.5 MB on-chip
Helium	MVE — halves the FFT
Compute share	**1.2 % of the energy cycle**
Battery life	1.50 y — *slightly worse* than the STM32U5

Branch `hw/alif-e7-ethos-u55`.

Never flashed. And the kit's reason for existing — the waveform model on the NPU — is **not written**. Writing it needs a 1D CNN trained on measured data, which no model in this repository has yet been.

A stub returning a plausible probability would make the kit look complete and would be the most misleading artefact here.

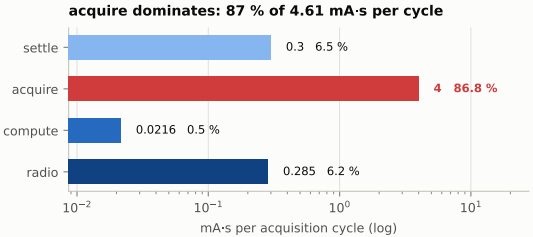
On the baseline pipeline, the NPU does nothing

	Latency	Share of the energy cycle
63-float detector, plain C	∼23 μs	0.011 %
CMSIS-NN on the M55	∼8 μs	—
Ethos-U55	∼2 μs	—
Whole DSP pipeline (Helium)	2.4 ms	1.2 %
Acquisition window	**307 ms**	**66.9 %**

An NPU that takes inference from 23 μs to 2 μs is optimising **0.010 %** of the budget. Helium, which halves the FFT, optimises 1.2 %.

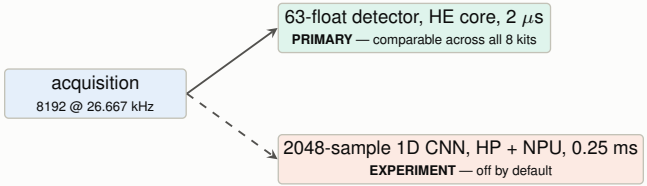
These were 0.0005 % and 0.5 % until the energy model's acquisition window was corrected from a flat 1000 ms to this firmware's 307 ms. The NPU figure moved by 20× and the conclusion did not — which says it was never resting on the number.

A node on this part has materially the same battery life as one on a 90 EUR STM32U5 — in fact *slightly worse*, because the sleep current is a little higher.



So what: Every "AI microcontroller" evaluation should start with this table. On a low-duty sensor the dominant term is the sensor, not the maths.

What it does buy: a model that does not otherwise fit



At ∼20 GMAC/s the small Ethos-U55 runs a 2–5 MMAC 1D CNN over the raw waveform in **0.25 ms**. The same model is 100–250 ms on the HE core alone and ∼600 ms on the nRF9160's 64 MHz M33 — longer than the acquisition it would be analysing.

That architecture is what the 2025–2026 literature actually uses: end-to-end 1D CNN on raw vibration, dual-domain multi-scale CNN at -4 dB SNR, lightweight on-device transformers. *None of them run on any other kit here.*

Why the experiment is off by default

- **Comparability.** Eight kits exist to be compared; a different default produces frames that cannot sit beside the others.
- **Energy.** Running it every cycle costs more than the whole rest of the cycle — and 99.99 % of acquisitions are on healthy machines.
- **Epistemics.** The result is reported *alongside* the feature frame. The interesting output is the **disagreement set**; replacing the comparable path destroys the only evidence that would settle which model is right.

So what: The NPU changes which hypothesis the node can test — not how fast it tests the current one.

Power domains matter more than TOPS; and one flag costs 2×

The HP core must be *absent*, not idle

	Current	Role
M55 HE, 160 MHz	4.5 mA	the whole normal cycle
M55 HP, 400 MHz	22 mA	**powered off**

22 mA left *idle* between hourly acquisitions costs 79 mA·s per hour — **17× the entire rest of the cycle**.

So the real argument for a heterogeneous SoC in a battery application is about *gating*, not throughput. A part whose big core cannot be fully powered down is worse here than a small one, whatever its TOPS figure.

`-mfpu=auto`

On a Cortex-M55 this flag is what enables MVE. Without it the compiler emits scalar code, CMSIS-DSP silently falls back to its generic paths, and the FFT takes twice as long.

It produces identical numbers. Golden vectors match. Edge/cloud feature agreement matches. Quantisation round-trip matches. *Every test in this repository passes either way.*

A 2× regression invisible to the whole test suite is exactly the kind of defect that ships. The only detection is a PMU cycle count on hardware — which is why it is on the checklist and why the flag carries a comment rather than just being set.

Same hazard on the NPU: Vela's operator fallbacks are silent, and one unsupported operator runs on the CPU at 1/100th the speed with the same answer.